

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split, GridSearchCV, cross_val_score
from sklearn.svm import SVC
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
data = load_breast_cancer()

X = data.data
y = data.target

print("Features shape:", X.shape)
print("Target classes:", data.target_names)
```

```
Features shape: (569, 30)
Target classes: ['malignant' 'benign']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
svm_linear = SVC(kernel='linear', C=1)
svm_linear.fit(X_train, y_train)

y_pred_linear = svm_linear.predict(X_test)

print("Linear Kernel Accuracy:",
      accuracy_score(y_test, y_pred_linear))
```

```
Linear Kernel Accuracy: 0.956140350877193
```

```
svm_rbf = SVC(kernel='rbf', C=1, gamma='scale')
svm_rbf.fit(X_train, y_train)

y_pred_rbf = svm_rbf.predict(X_test)

print("RBF Kernel Accuracy:",
      accuracy_score(y_test, y_pred_rbf))
```

```
RBF Kernel Accuracy: 0.9824561403508771
```

```
param_grid = {
    'C': [0.1, 1, 10, 100],
    'gamma': [1, 0.1, 0.01, 0.001],
    'kernel': ['rbf']
}

grid = GridSearchCV(SVC(), param_grid, refit=True, verbose=2, cv=5)
grid.fit(X_train, y_train)
```

```
print("Best Parameters:", grid.best_params_)
```

```
Fitting 5 folds for each of 16 candidates, totalling 80 fits
[CV] END .....C=0.1, gamma=1, kernel=rbf; total time= 0.0s
[CV] END .....C=0.1, gamma=1, kernel=rbf; total time= 0.0s
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[CV] END .....C=0.1, gamma=0.001, kernel=rbf; total time= 0.0s
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[CV] END .....C=0.1, gamma=0.001, kernel=rbf; total time= 0.0s
[CV] END .....C=0.1, gamma=0.001, kernel=rbf; total time= 0.0s
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[CV] END .....C=1, gamma=1, kernel=rbf; total time= 0.1s
[CV] END .....C=1, gamma=1, kernel=rbf; total time= 0.0s
[CV] END .....C=1, gamma=1, kernel=rbf; total time= 0.0s
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[CV] END .....C=10, gamma=1, kernel=rbf; total time= 0.1s
[CV] END .....C=10, gamma=1, kernel=rbf; total time= 0.0s
[CV] END .....C=10, gamma=1, kernel=rbf; total time= 0.0s
[CV] END .....C=10, gamma=1, kernel=rbf; total time= 0.1s
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[CV] END .....C=10, gamma=0.01, kernel=rbf; total time= 0.0s
```

```
scores = cross_val_score(SVC(kernel='linear'), X, y, cv=5)
print("Cross-validation accuracy:", scores.mean())
```

Cross-validation accuracy: 0.9455364073901569